

CS168

Introduction to the Internet: Architecture and Protocols

Sylvia Ratnasamy and Ion Stoica
Fall 2026

Today

- **Introductions**
- **Class logistics**
- **What is (this course on) the Internet about?**

Ion Stoica (he/him)

- **Background**

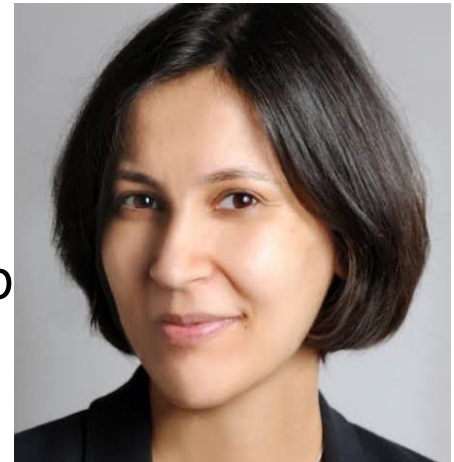
- PhD from Carnegie Mellon University
- Joined the UCB faculty in 2001
- Projects: Apache Spark, Ray, SkyPilot, vLLM, SGLang, SkyRL, ...
- Co-founder: Databricks, Anyscale, Conviva
- Research focus: AI and Cloud Computing



Sylvia Ratnasamy (she/her)

- **Background**

- PhD from UC Berkeley
- Joined the UCB faculty in 2011
- Industry experience: Intel; co-founded a startup
- Research focus: networks and systems

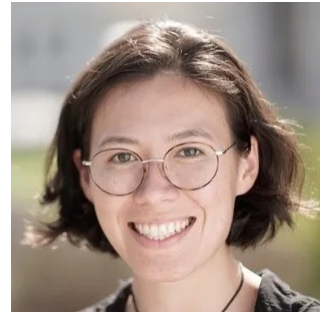


TAs!

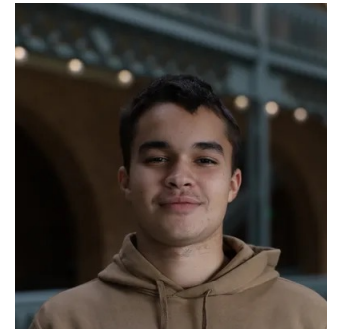
**Zoir
Imomaliev
(head TA)**



**Abhi
Nambiar**



**Tess
Despres**



Mehdi Khfifi



Sam Son

Tutors!



Sydney Dinh

Arnav



**Riya
Sehgal**

Valerie Eng

Today

- Introductions
- **Class logistics**
- **What is (this course on) the Internet about?**

Class Logistics

- All course logistics are available on our website:
<https://fa26.cs168.io/>
- Policies page: <https://fa26.cs168.io/policies/>
- FAQs page: <https://fa26.cs168.io/fa26-faqs/>
- If you have questions, please read our FAQs page first!
- If your question is not answered on that page, please email cs168@berkeley.edu with your question. Do not email individual instructors or TAs; it is far less likely that you will get a response

Enrollment, Ed, bCourses, & Gradescope

- Please do not email us if you are a concurrent enrollment student with a pending application; subject to the enrollment caps, we will add you automatically within 3–4 days of submitting your application.
- If you just enrolled in the class, please don't email us about being added; subject to the enrollment caps, we will sync the roster and add you within 3–4 days

Discussions, Office Hours, and Exams

- Discussions and office hours start next week.
- You can attend any discussion section. Attendance is not taken.
- Project 1 starts today and the deadline is September 06
- Exam dates and details about alternate exams are on the website.

Accommodations

- If you are registered with the Disabled Students' Program (DSP), please send us your letter of accommodations through the DSP portal as soon as possible.
- Your well-being is more important than this class. The website has a link to a form to request extensions.

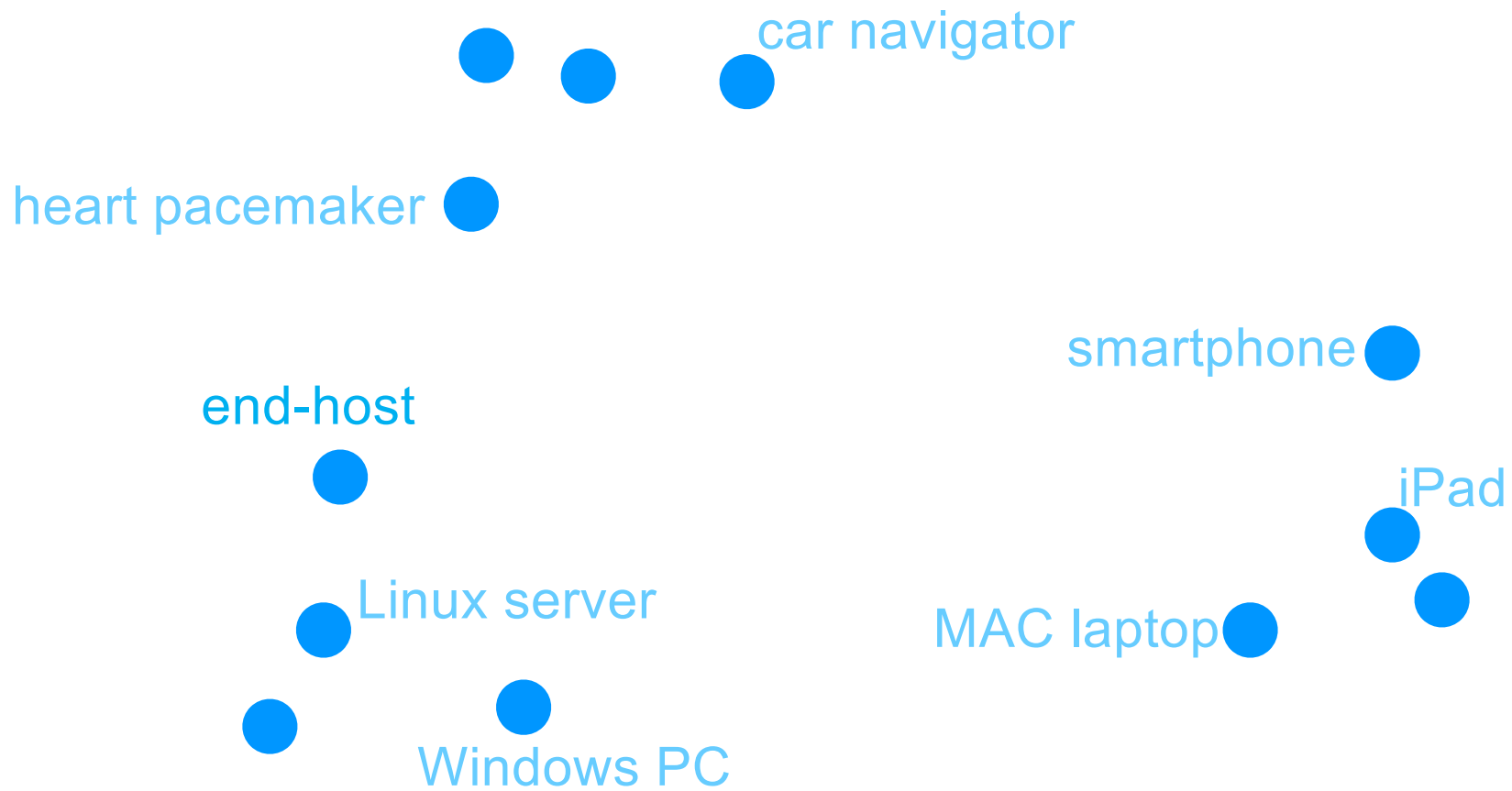
Today

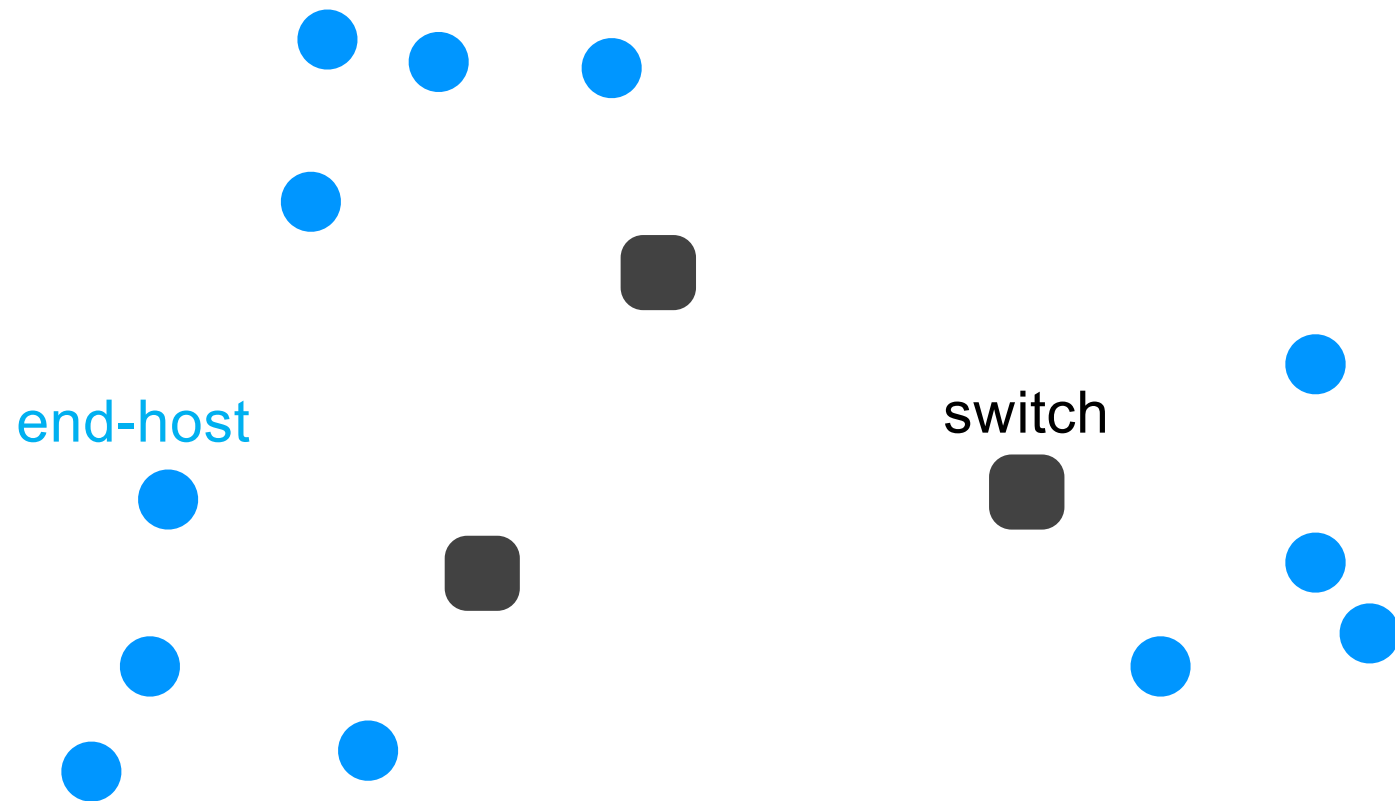
- Introductions
- Class logistics
- **What is (this course on) the Internet about?**

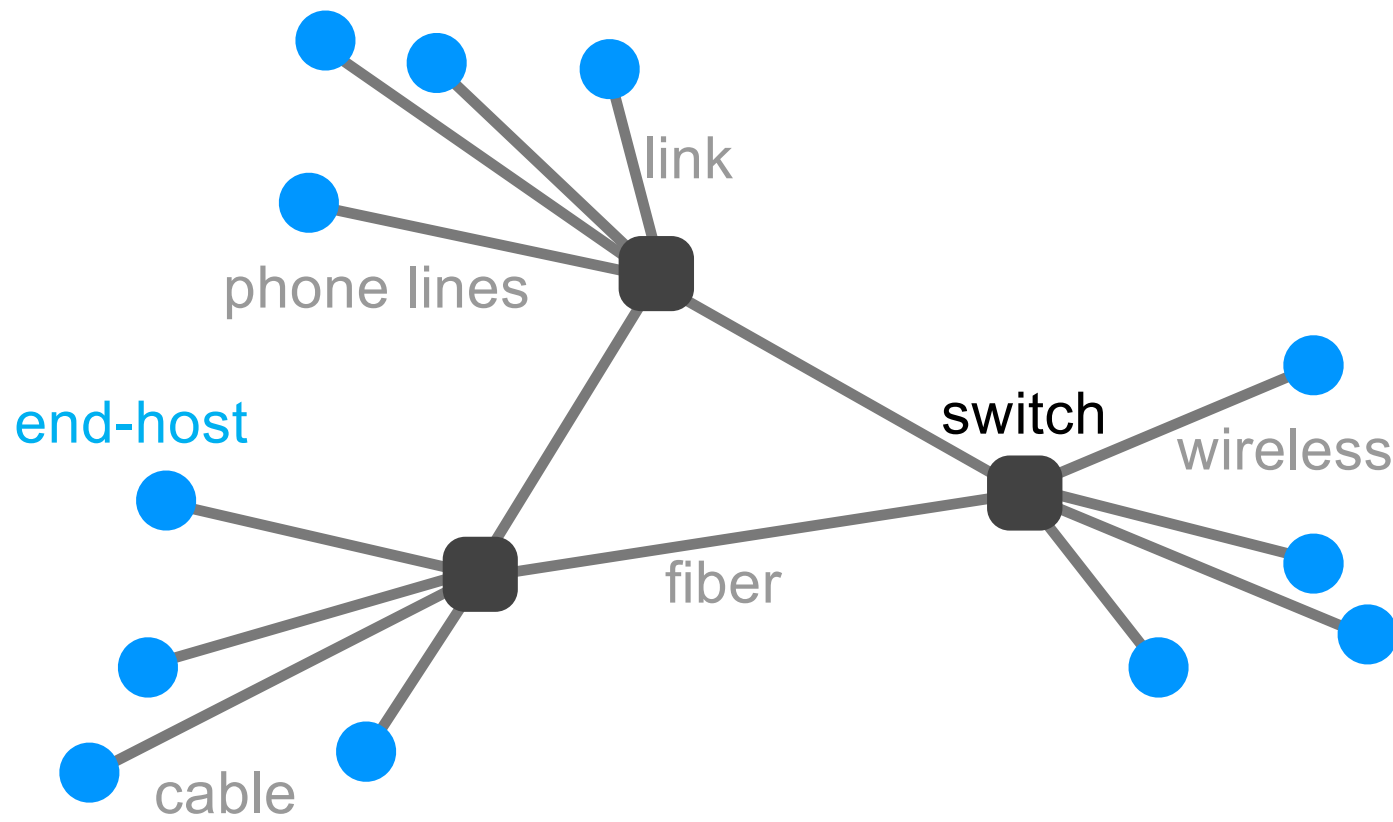
- **Internet**
- **Protocols**
- **Architecture**

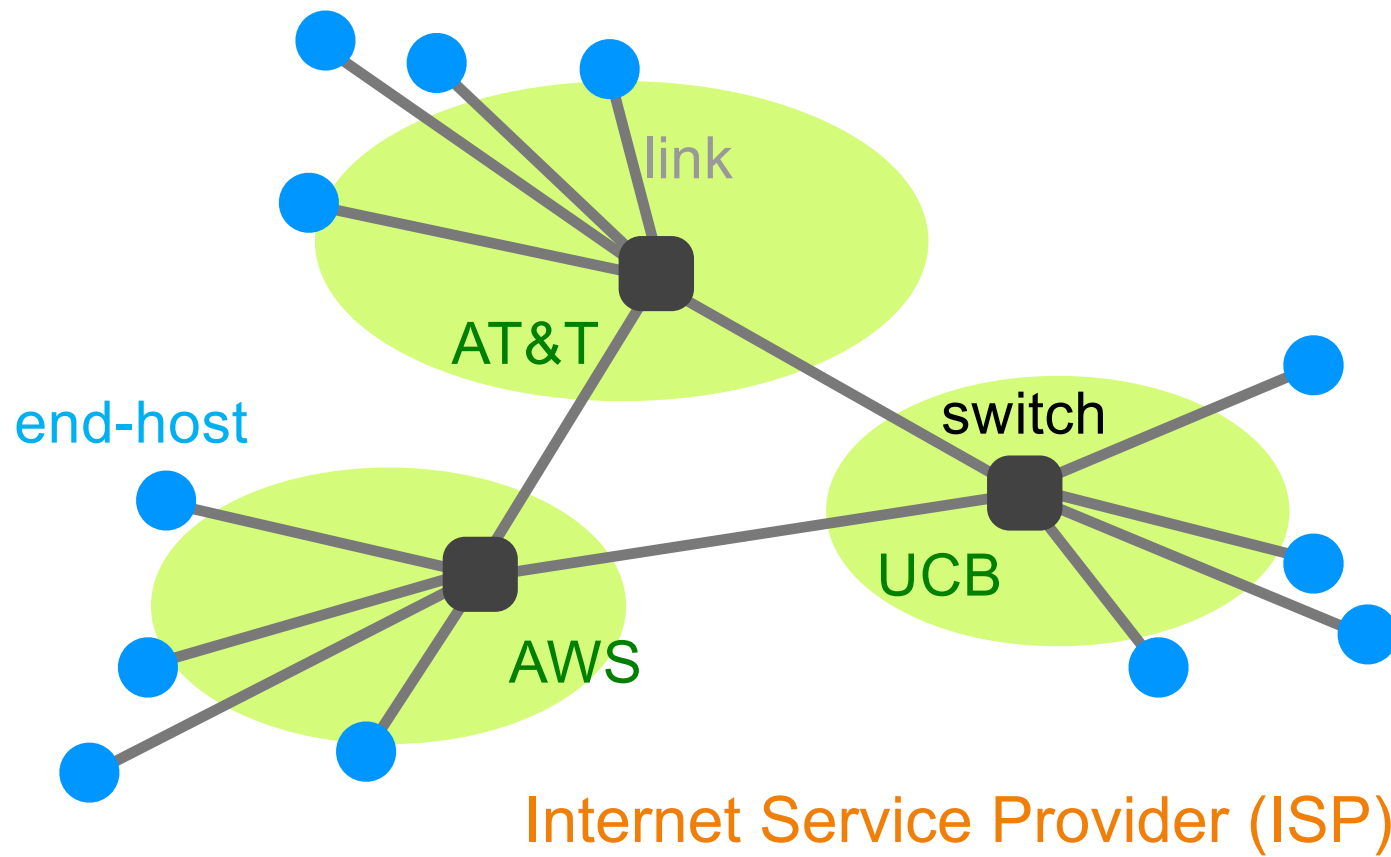
Two Meanings of “Internet”

- **The infrastructure that ties together computing devices**
 - TCP, IP, BGP, DNS, OSPF, ...
- **The ecosystem of applications built on top of the above infrastructure**
 - amazon, twitter, chatGPT
- **In this class, we use the first definition!**

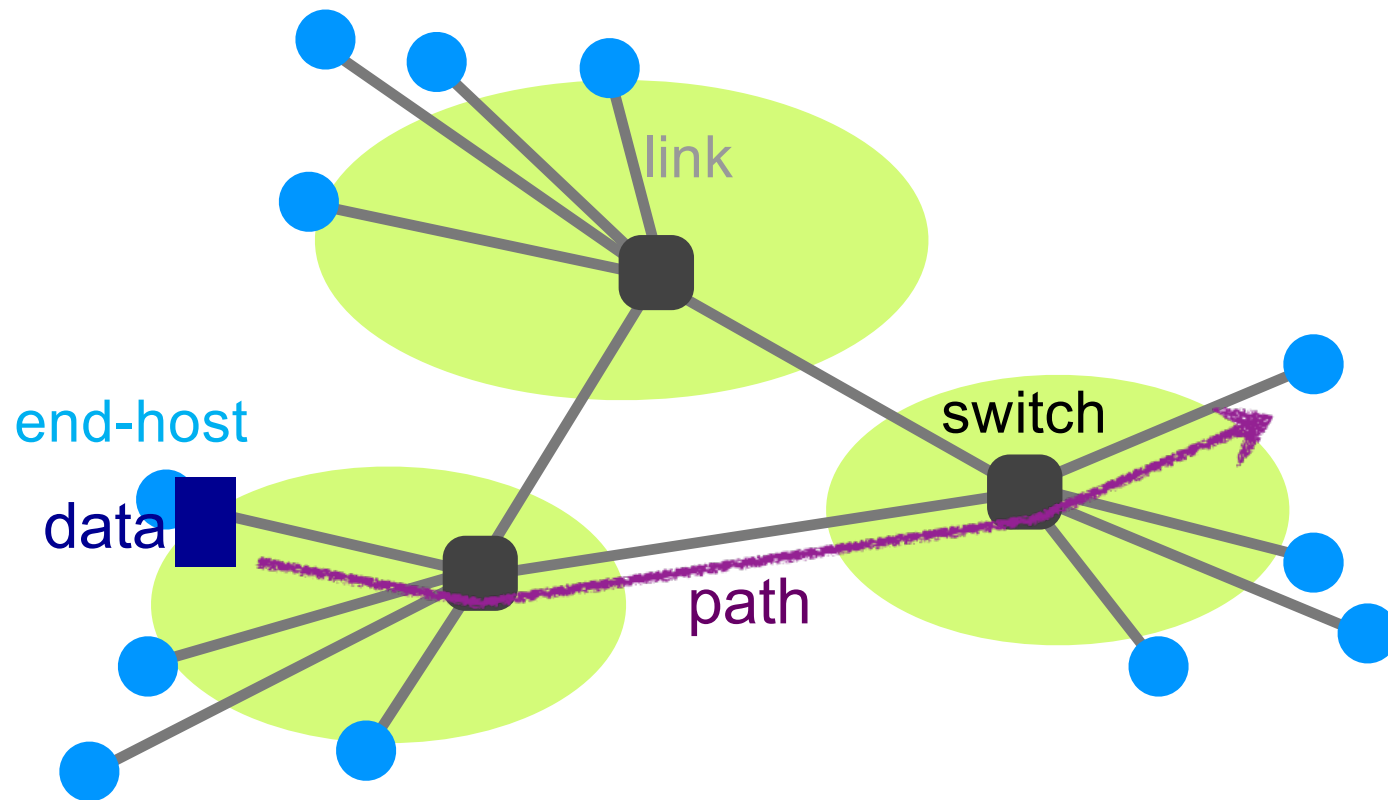








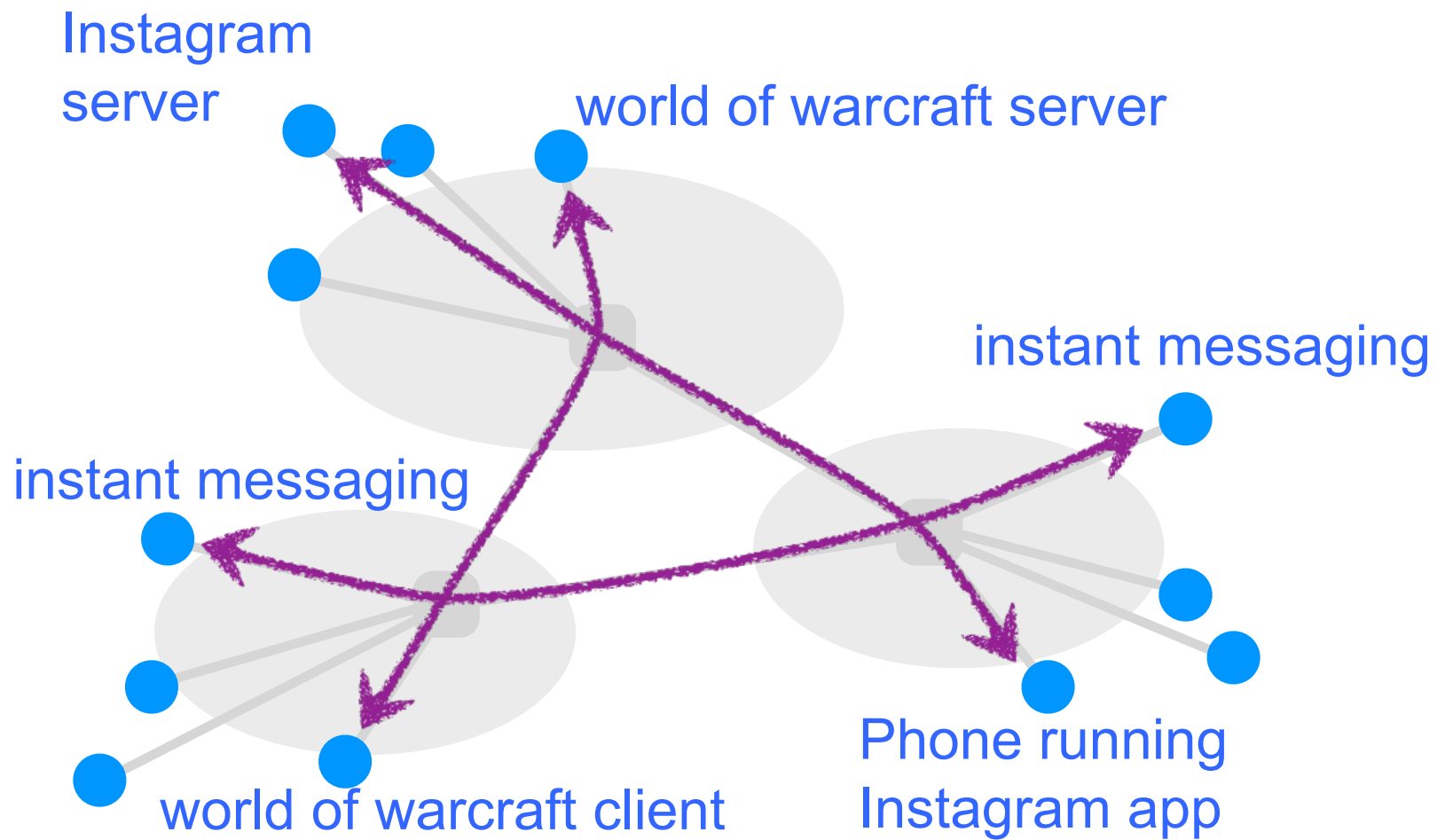
The Internet transfers data between end hosts

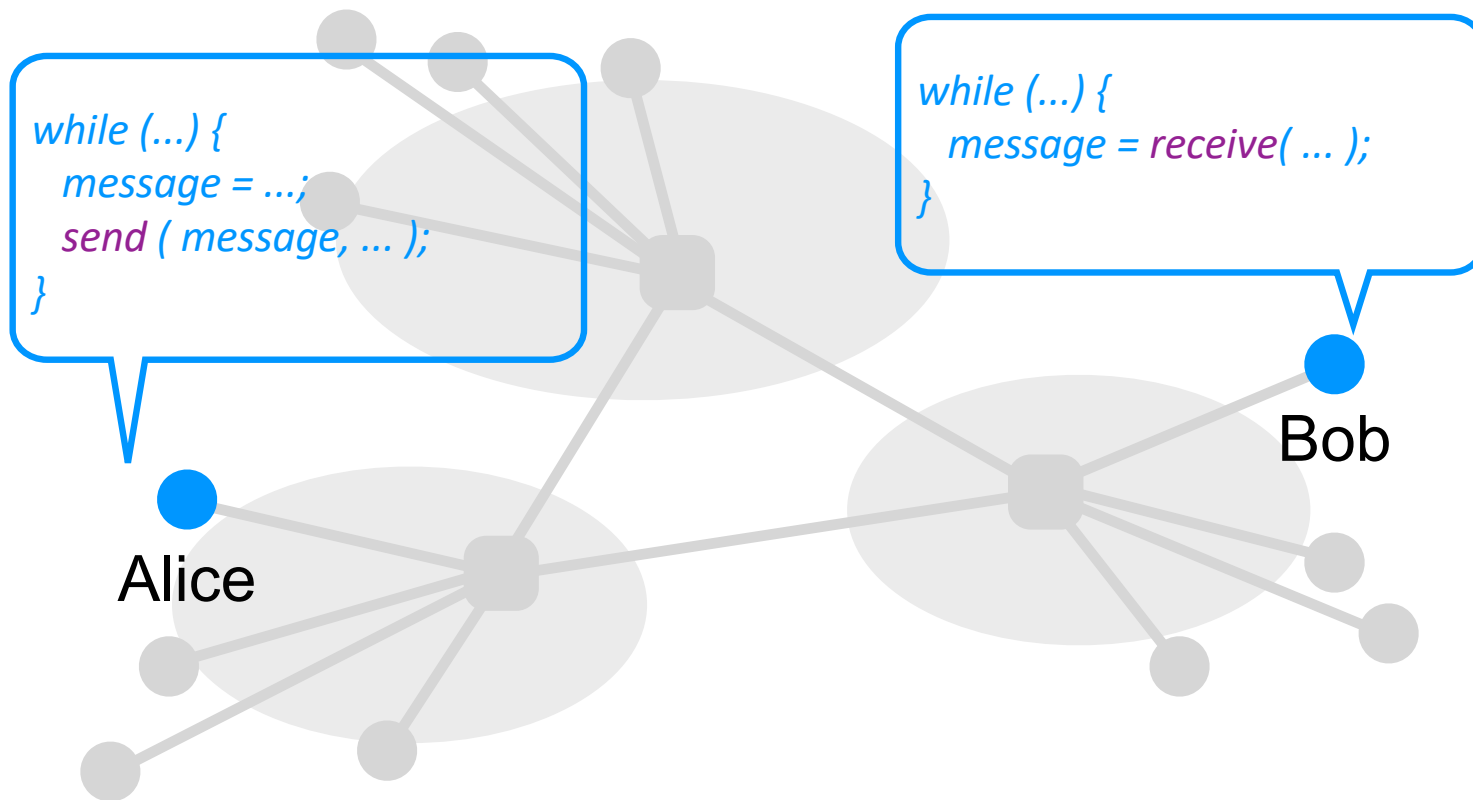


- Internet

- **Protocols**

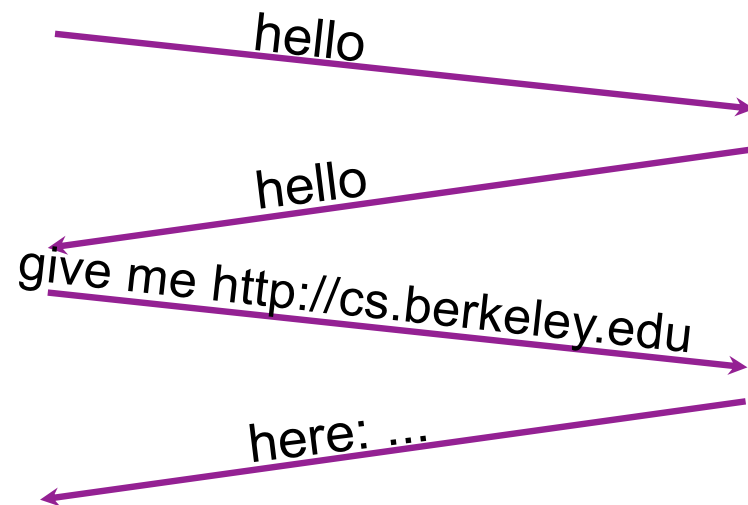
- **Architecture**





Alice

Bob



Alice

Bob



Protocol

- A specification of the messages that communicating entities exchange
 - their syntax and semantics
- Very much like conversational conventions ... determining who should talk next and how they should respond
- Designing a good protocol is harder than it first seems!

- Internet

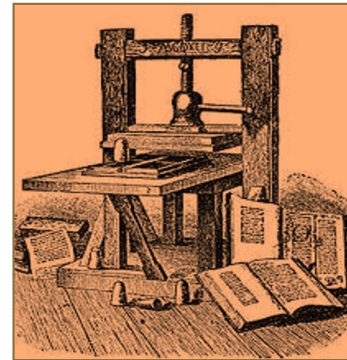
- Protocols

- **Architecture**

Why study the Internet?

The Internet has and is transforming everything

- **The way we do business ...**
 - retail, advertising, cloud computing
- **The way we have relationships**
 - Twitter, chat
- **The way we learn**
 - Wikipedia, ChatGPT, AR/VR
- **The way we govern**
 - E-voting, censorship, cyber-warfare
- **The way we cure disease**
 - digital health, remote surgery



What's your formal model for the Internet? -- theorists

You don't have performance benchmarks??? – hardware folks

But why is the Internet *interesting*?

Aren't you just writing software for networks? – OS community

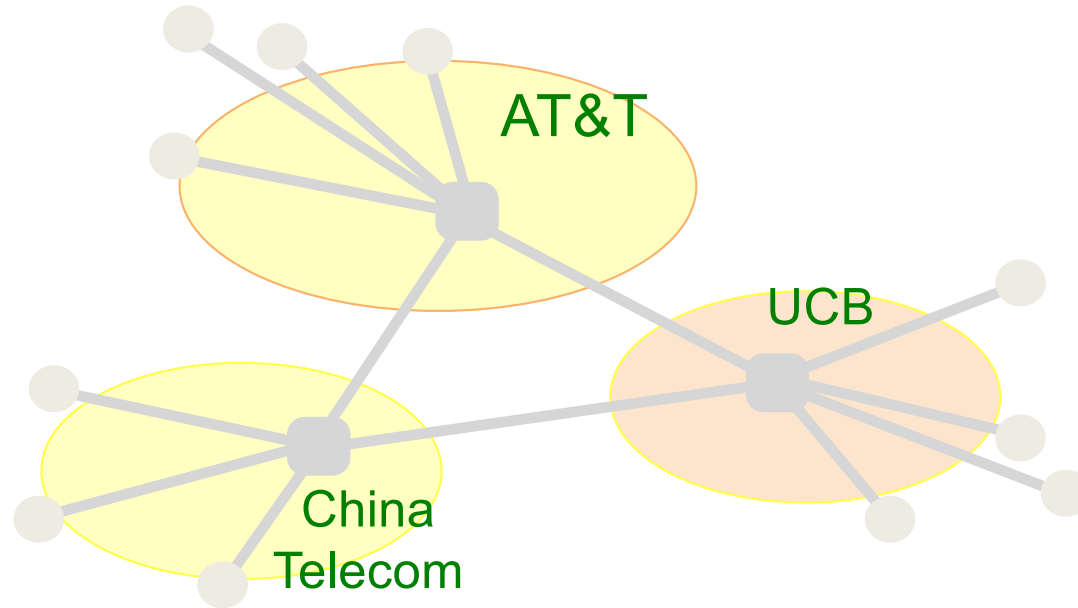
A few defining characteristics of the Internet...

Network versus “The Internet”

- **There are many kinds of network technologies (switches and links)**
 - Ethernet, optical, wifi access points, DSL modems, Infiniband switches, ...
- **The Internet is not a new/particular kind of network technology**
- **Instead, the Internet ties different networks together**
 - The Internet

A federated system

Interoperability is the Internet's most important goal!



The Internet interconnects over 100,000 independently operated networks

A federated system

- **Fundamental challenge: how do you interconnect competing entities?**
 - Competing network providers must cooperate to serve their customers!
- **Leads to a constant tussle between business and technical factors**
 - Real-world incentives determine topology, path selection, diagnostics, and more
- **And complicates innovation**
 - How do you differentiate when interoperability relies on supporting a common protocol?
 - Upgrading “the Internet” is not an option

Tremendous scale

- > 5 Billion users
- 1.24 Trillion unique URLs (web pages)
- Every second, we generate >10000 tweets, >100,000 Google queries, >3M emails

Asynchronous Operation

- Fundamental constraint: **speed of light**
- Consider: how many cycles does your 3GHz CPU in Berkeley execute before it can possibly get a response for a message it sends to a server in NY?
 - Berkeley to New York: 4,125 km
 - Traveling to NY and back at 300,000 km/s: 27.5 milliseconds
 - $3,000,000,000 \text{ cycles/sec} * 0.0275 = 84,000,000 \text{ cycles!}$
- Thus, communication feedback is always **dated**

Prone to Failure

- Many components along a path
 - software, switches, links, network interface cards, wireless access points, modem,...
- Consider: 50 components, that work correctly 99% of time → 39.5% chance communication fail
 - Plus asynchrony → takes a long time to hear (bad) news

Enormous diversity and dynamic range

- **Technologies:** optical, wireless, satellite, copper, ...
- **Communication latency:** microseconds to seconds (10^6 operating range)
- **Bandwidth:** 1Kbits/second to 1 Terabit/second (10^8 operating range)
- **Reliability:** 0 – 90%
- **Devices:** sensors, cell phones, datacenters, ...
- **Users:** you and me, governments, operators, malicious users, ...
- **Applications:** WhatsApp, live video, gaming, remote medicine, ...

Constant evolution

1970s:

- 10^4 bits/second links
- < 100 computers in the US
- File transfer is the “killer” app

Today

- 10^{14} bits/second links
- 10B+ devices, all over the globe
- 3B+ facebook users; self-driving cars

Yet change must be backward compatible, incremental, and “in place”

Recap: The Internet is ...

- A federated system ...
- of enormous scale ...
- with tremendous dynamic range and diversity ...
- that is asynchronous in operation ...
- failure prone ...
- **and constantly evolving**

Recap: The Internet is ...

- Too complex for theoretical models
- “Working code” needn’t mean much
- Performance benchmarks are too narrow

The creation of the Internet required a new design paradigm

The Internet design paradigm

- Decentralized control
- A best-effort service model
- "Route around trouble"
- Dumb infrastructure (w/ smart endhosts)
- The end-to-end design principle
- Layering
- Federation via a "narrow waist" interface

**A radical departure from systems at
the time**

Example: a best-effort service model

- **Fundamental question: what's the right service model that a network should support?**
 - "contract" between network and its users/end-hosts
- **Some possibilities:**
 - "guarantee that data will be delivered"
 - "guarantee that data will be delivered within X time"
 - "return a confirmation of successful delivery or an error"
- **Instead, what the Internet supports: "best effort" delivery of data**
 - No guarantee on whether or when data will be delivered
 - No notification of outcome!

The Internet design paradigm

- Decentralized control
- A best-effort service model
- “Route around trouble”
- Dumb infrastructure (w/ smart endpoints)
- The end-to-end design principle
- Layering
- Federation via a “narrow waist” interface

A radical departure from systems at the time
Now the de-facto blueprint for scalable services

The Internet design paradigm

- Decentralized control → **SDN: centralize?**
- A best-effort service model → **Lossless fabrics for AI clusters**
- “Route around trouble”
- Dumb infrastructure (w/ smart endpoints) → **in-network storage for LEO networks**
- The end-to-end design principle → **Edge computing?**
- Layering → **cross-layer optimizations**
- Federation via a “narrow waist” interface

But it is just one design ...

... that is *constantly* being questioned

Backing up a level

- **The Internet poses a design challenge like no other**
- **From its creation emerged a new design paradigm**
- **That shaped how we reason about the design of scalable systems**
 - What's the right prioritization of goals?
 - What are fundamental constraints?
 - How do we decompose a problem?
 - What abstractions do we need?
 - What are the tradeoffs?
- **In short, a lesson in how to architect a (networked) system**

- Internet

- Protocols

- **Architecture**

Network architecture

- More about thinking rigorously than doing rigorous math
- More about understanding tradeoffs than running benchmarks
- More about practicality than optimality

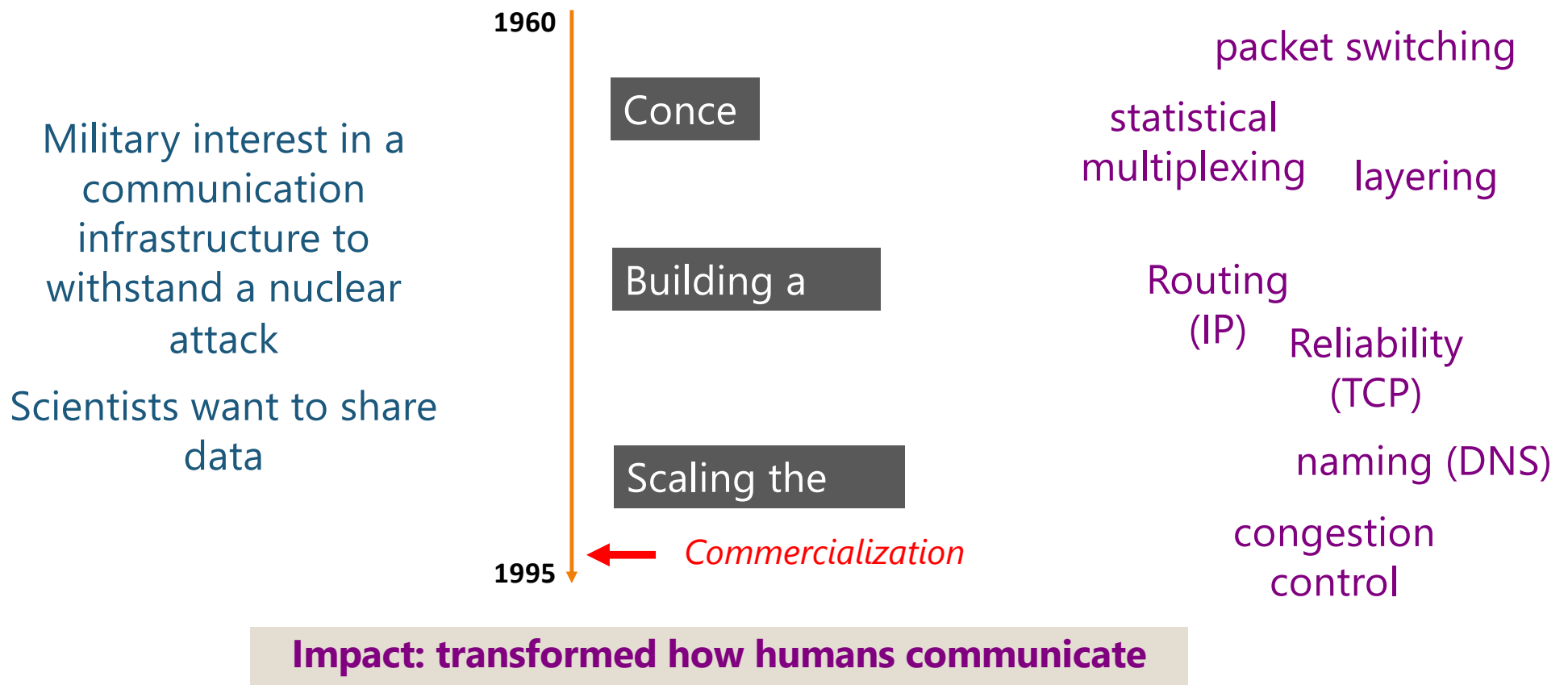
Done right, can be a powerful thing!

Class topics, more concretely

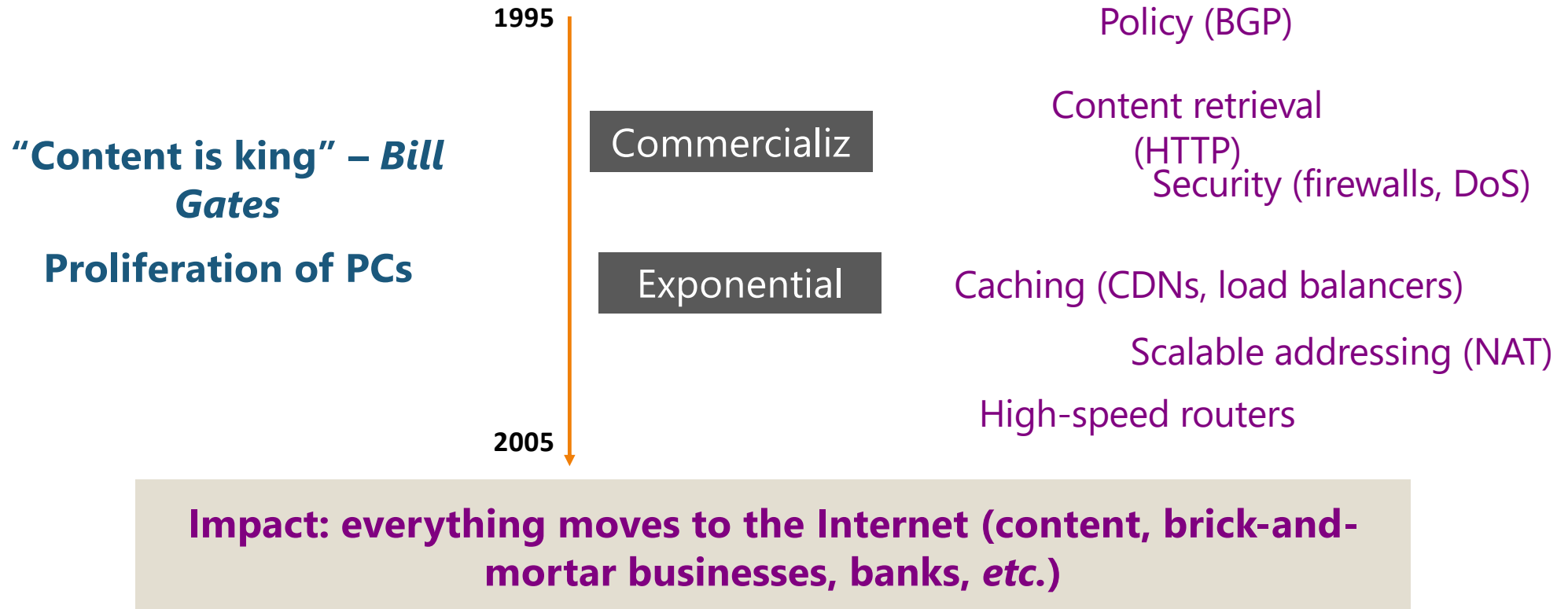
Reflect three broad phases in the Internet's evolution

1. Building a global data communication network
2. Scaling communication; and the emergence of a commercial ecosystem
3. (Networks that enable) scaling data; and a shifting commercial ecosystem

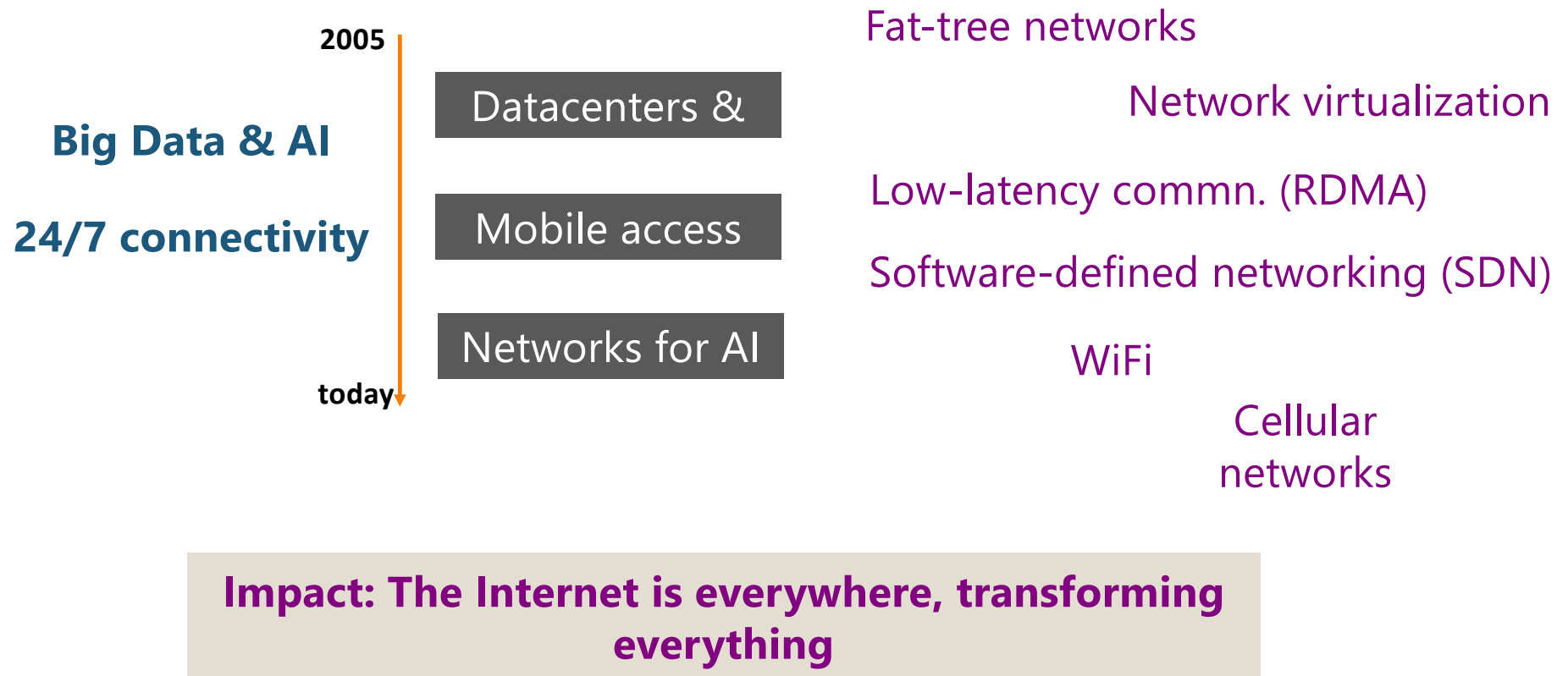
Phase 1: Building a global data communication network



Phase 2: Scaling & the emergence of a commercial ecosystem



Phase 3: Data and a shifting ecosystem



To recap, what we hope CS 168 will teach you

- **How the Internet works**
- **Why it works the way it does**
- **How to reason through a complex (networking) design problem**